

Jay Yoon

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<https://jayakajaeha.github.io>

EDUCATION

Sejong University

B.Eng. in Artificial Intelligence (Expected Feb. 2027)

- GPA: 4.38/4.5

Gwangjin, S. Korea

Mar. 2021 – Feb. 2027

RESEARCH EXPERIENCE

Music and Audio Research Group (MARG)

Research Intern, Advisor: Kyogu Lee

Seoul National University

Jul. 2026 – Aug. 2026

- Analyzed text-token attention and instrument-related activations in MusicGen, identifying early signals associated with violin omission.
- Tested the causal role of initial tokens through transplantation experiments, showing that replacing early tokens could restore violin generation.
- Compared attention interventions, activation steering, and classifier-free guidance as mitigation strategies, reducing omission from 14.4% to 1.4% through guidance adjustment.

Pattern Recognition & Machine Learning Lab

Undergraduate Researcher, Advisor: Chang-Jae Chun

Sejong University

Jan. 2024 – Present

- Developed an explainable stacking ensemble for protein–protein docking quality assessment, combining multicollinearity filtering with SHAP-based feature selection.
- Trained on CASP15 and evaluated on CASP16, ranking 1st in both SCORE and QSCORE.
- Preparing a first-author journal manuscript and participating in CASP17.

PUBLICATIONS

- **Yoon, J.**, Kim, N., Kim, D., & Chun, C.-J. "A Multi-level Stacking Ensemble for Protein Structure Quality Prediction." KICS Winter Conf., 2026.
- Hwang, S., **Yoon, J.**, & Chun, C.-J. "Ensemble-Based Quality Assessment Model for Protein Docking Candidate Structures." KICS Winter Conf., 2026.
- Kim, D., **Yoon, J.**, & Chun, C.-J. "Efficiency Analysis of LHS-Based Sampling Across Different Data Usage Ratios." KICS Winter Conf., 2026.
- Lee, D., **Yoon, J.**, Kim, S., Lee, S., Min, B., & Kim, J. "Semantic Decomposition based Explainable Anomaly Video Retrieval System." KIBME Summer Conf., 2026.

SELECTED PROJECTS

POCHAK: Multimodal Intelligent Video Retrieval System **Sejong University**
Capstone Design Project (Grand Prize) **Mar. – Jun. 2026**

- Developed a multimodal video retrieval system that searches large-scale CCTV footage using natural language queries, image cues, and person Re-ID.
- Designed a three-level semantic captioning pipeline that decomposes visual scenes into background, object, and situation-focused representations.

DIVO: AI Phone Reservation System for Demand-Responsive Transit **KT**
AI Developer (Best Project Award) **May – Aug. 2026**

- Fine-tuned Whisper Large-v3 using LoRA for telephone speech recognition, reducing recognition error from 11.1% to 5.9%.
- Contributed to a reservation pipeline combining speech recognition, natural language understanding, rule-based validation, and dialogue state management.

Jongno-gu Public Assembly and Traffic Alert Service **KT & Jongno-gu Office**
Project Developer **Jul. – Aug. 2025**

- Developed a notification chatbot adopted by Jongno-gu Office for public assembly alerts; received a commendation from the district mayor.
- Built an automated pipeline combining bus route data collection, LLM-based extraction of traffic control information, and a notification API for real-time traffic alerts.

SELECTED HONORS & AWARDS

KT Digital Talent Scholarship	Mar. 2025 – Present
KT Digital Talent Award – “Best Project Award”	Aug. 2026
Grand Prize, Sejong University Creative Design Competition	Jun. 2026
Best Prize, Sejong AI Challenge	Jun. 2026